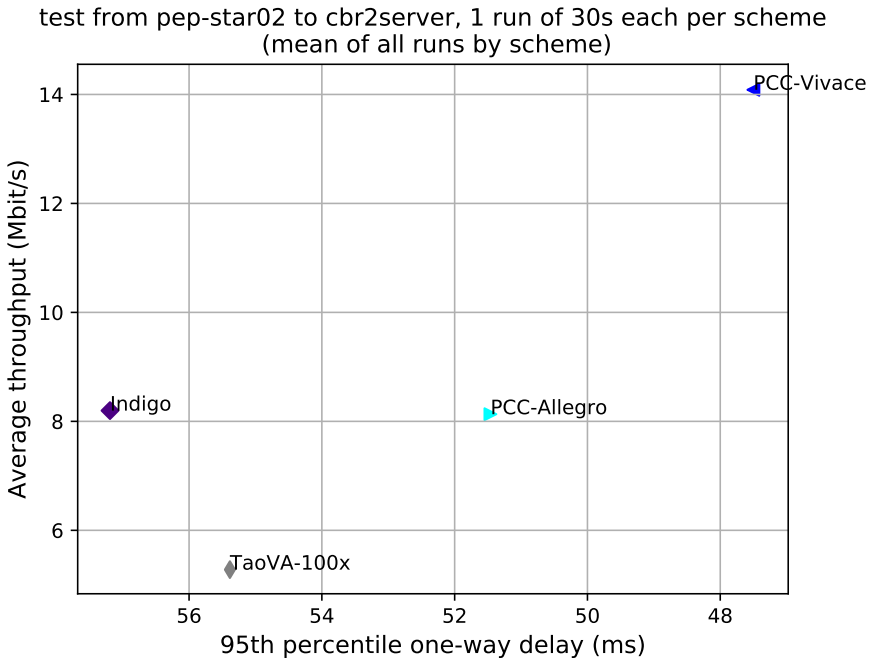
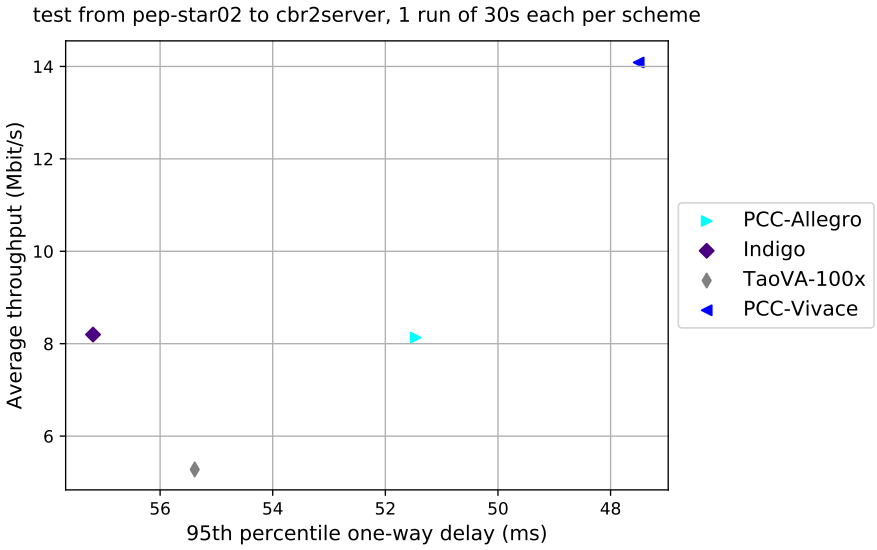
Pantheon Report

Generated at 2026-11-09 23:39:41 (UTC).
Data path: pep-star02 (*local*) →cbr2server (*remote*).
Repeated the test of 4 congestion control schemes once.
Each test lasted for 30 seconds running 1 flow.
NTP offsets were measured against 202.65.114.202 and have been applied to correct the timestamps in logs.

```
System info:
Linux 7.0.0-42-generic
net.core.default_qdisc = fq
net.core.rmem_default = 212992
net.core.rmem_max = 4194304
net.core.wmem_default = 212992
net.core.wmem_max = 4194304
net.ipv4.tcp_rmem = 4096 131072 33554432
net.ipv4.tcp_wmem = 4096 16384 4194304

Git summary:
branch: HEAD @ 23e738ce5acae1d36e321886cd613b0b9401ac11
third_party/genericCC @ d0153f8e594aa89e93b032143cedbdfe58e562f4
third_party/indigo @ 463d89b09699a57bfdfbae351646df6a60040b90
third_party/pantheon-tunnel @ f866d3f58d27afd942717625ee3a354cc2e802bd
third_party/pcc @ 1afc958fa0d66d18b623c091a55fec872b4981e1
 M receiver/src/buffer.h
 M receiver/src/core.cpp
 M sender/src/buffer.h
 M sender/src/core.cpp
third_party/vivace @ 2baf86211435ae071a32f96b7d8c504587f5d7f4
```


test from pep-star02 to cbr2server, 1 run of 30s each per scheme
(mean of all runs by scheme)
Average throughput (Mbit/s)
95th percentile one-way delay (ms)
PCC-Vivace
Indigo
PCC-Allegro
TaoVA-100x
14
12
10
8
6
56
54
52
50
48


test from pep-star02 to cbr2server, 1 run of 30s each per scheme
Average throughput (Mbit/s)
95th percentile one-way delay (ms)
PCC-Allegro
Indigo
TaoVA-100x
PCC-Vivace
14
12
10
8
6
56
54
52
50
48

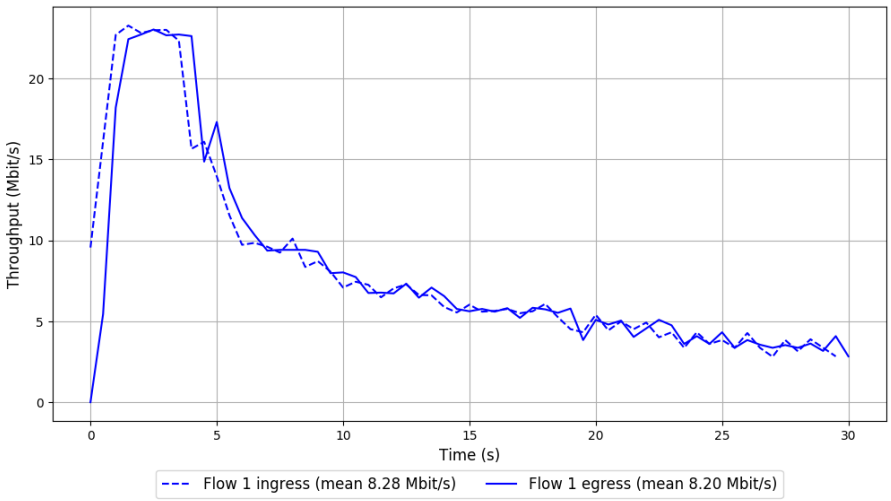
scheme	# runs	mean avg tput (Mbit/s) flow 1	mean 95th-%ile delay (ms) flow 1	mean loss rate (%) flow 1
Indigo	1	8.20	57.19	0.89
PCC-Allegro	1	8.13	51.46	0.00
TaoVA-100x	1	5.28	55.38	0.79
PCC-Vivace	1	14.09	47.50	0.15

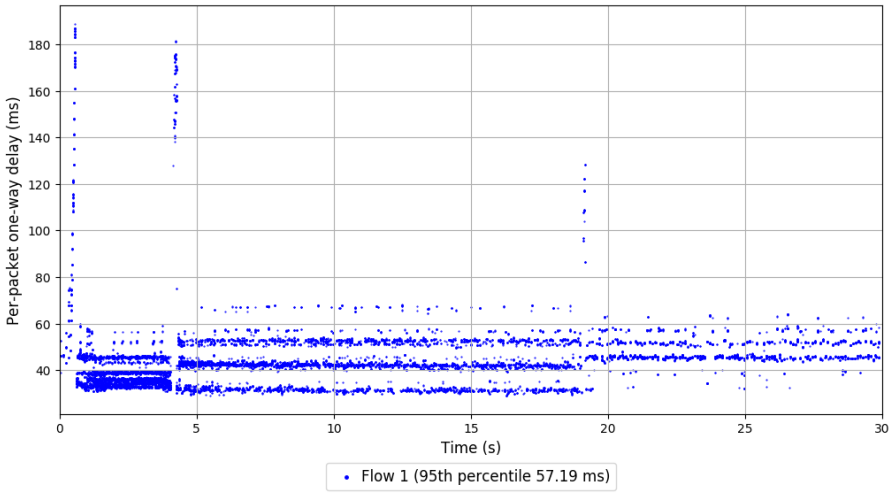
```
Run 1: Statistics of Indigo

Start at: 2026-11-09 22:17:07
End at: 2026-11-09 22:17:37
Local clock offset: 1.817 ms
Remote clock offset: -1.548 ms

# Below is generated by plot.py at 2026-11-09 23:39:40
# Datalink statistics
-- Total of 1 flow:
Average throughput: 8.20 Mbit/s
95th percentile per-packet one-way delay: 57.191 ms
Loss rate: 0.89%
-- Flow 1:
Average throughput: 8.20 Mbit/s
95th percentile per-packet one-way delay: 57.191 ms
Loss rate: 0.89%
```

Run 1: Report of Indigo — Data Link



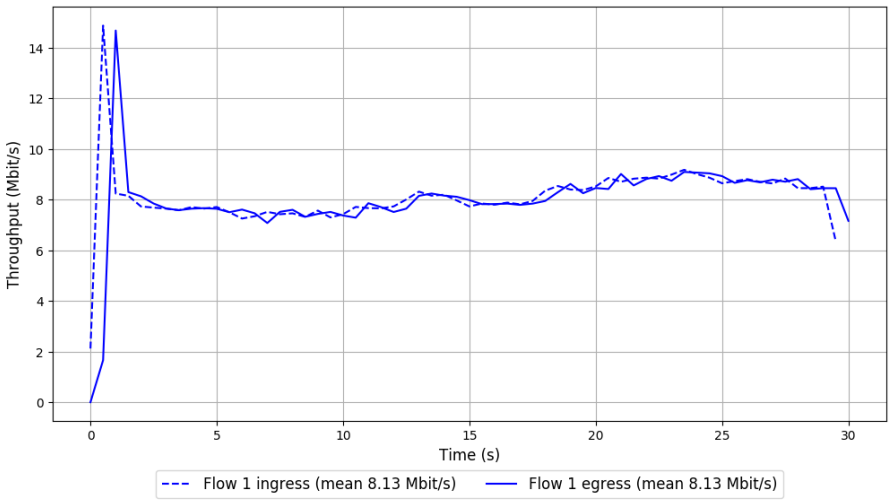


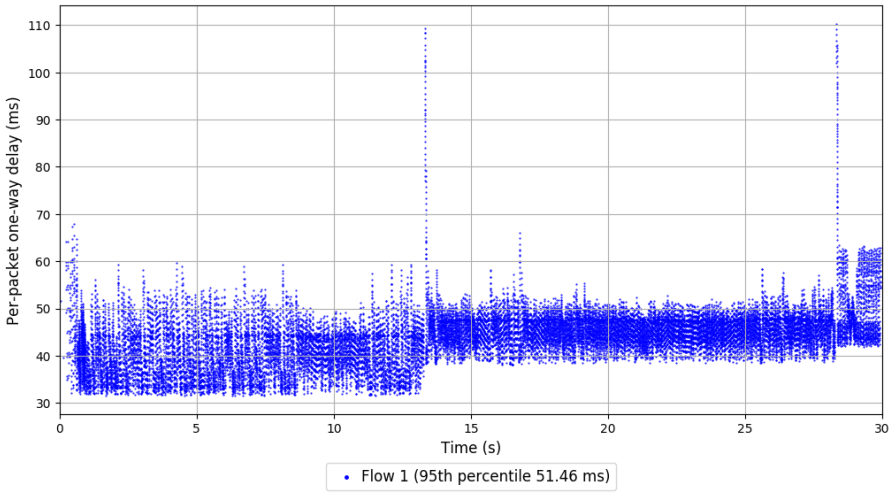
```
Run 1: Statistics of PCC-Allegro

Start at: 2026-11-09 22:15:58
End at: 2026-11-09 22:16:28
Local clock offset: 0.699 ms
Remote clock offset: -0.764 ms

# Below is generated by plot.py at 2026-11-09 23:39:40
# Datalink statistics
-- Total of 1 flow:
Average throughput: 8.13 Mbit/s
95th percentile per-packet one-way delay: 51.461 ms
Loss rate: 0.00%
-- Flow 1:
Average throughput: 8.13 Mbit/s
95th percentile per-packet one-way delay: 51.461 ms
Loss rate: 0.00%
```

Run 1: Report of PCC-Allegro — Data Link



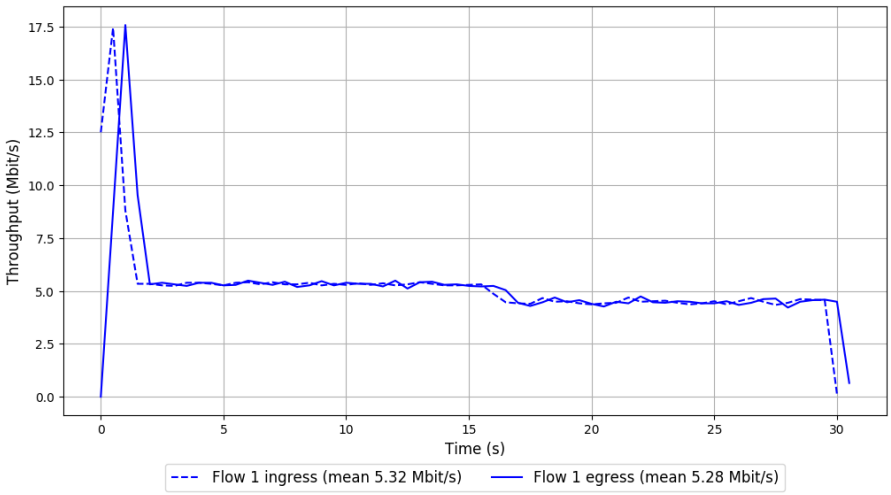


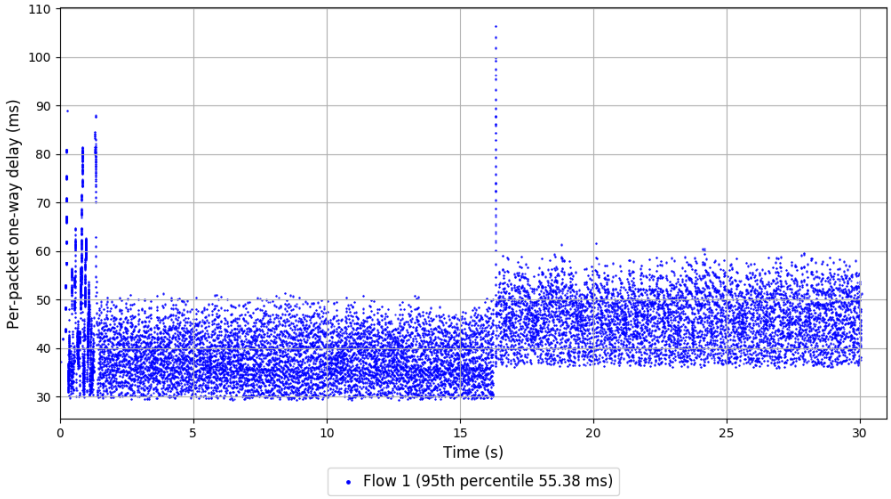
```
Run 1: Statistics of TaoVA-100x

Start at: 2026-11-09 22:19:25
End at: 2026-11-09 22:19:55
Local clock offset: 1.881 ms
Remote clock offset: -0.0 ms

# Below is generated by plot.py at 2026-11-09 23:39:40
# Datalink statistics
-- Total of 1 flow:
Average throughput: 5.28 Mbit/s
95th percentile per-packet one-way delay: 55.385 ms
Loss rate: 0.79%
-- Flow 1:
Average throughput: 5.28 Mbit/s
95th percentile per-packet one-way delay: 55.385 ms
Loss rate: 0.79%
```

Run 1: Report of TaoVA-100x — Data Link





```
Run 1: Statistics of PCC-Vivace

Start at: 2026-11-09 22:18:16
End at: 2026-11-09 22:18:46
Local clock offset: 0.784 ms
Remote clock offset: -0.138 ms

# Below is generated by plot.py at 2026-11-09 23:39:41
# Datalink statistics
-- Total of 1 flow:
Average throughput: 14.09 Mbit/s
95th percentile per-packet one-way delay: 47.501 ms
Loss rate: 0.15%
-- Flow 1:
Average throughput: 14.09 Mbit/s
95th percentile per-packet one-way delay: 47.501 ms
Loss rate: 0.15%
```

Run 1: Report of PCC-Vivace — Data Link

